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Initial post

by Sarra Fadhl Mohamed Binhadda Alsuwaidi - Thursday, 30 April 2026, 11:45 AM

Industry 4.0 and the evolving Industry 5.0 will be transformative of industries because of the new technology that has been introduced including AI, IoT and Data Analytics. An emphasis was placed on automation and efficiency by Metcalf (2024) when he described Industry 4.0; whereas an emphasis is now being placed on a human centric and resilient approach, from Metcalf (2024) in describing Industry 5.0. However, because of the growing dependency on digital systems the potential for failure increases. One notable example is the NHS IT System Outage of 2022. This outage disrupted patient care, delayed treatment for patients, decreased public confidence in digital health care. In addition to the financial loss associated with operational disruptions and restoration activities, there were also reputational consequences. When viewed from the perspective of Industry 5.0, this outage demonstrates the necessity for organizations to have organizational resilience, human oversight, and system designs that are capable of supporting these features. While the digital transformation enhances both efficiency and quality of service; organizations need to develop strategies that will allow them to manage risks and provide reliable digital systems so they can operate sustainably and securely.

Reference

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Peer Response

by Tamim Al-Mutawa - Sunday, 10 May 2026, 10:57 AM

Hello Sarra,

Your discussion clearly explains how Industry 4.0 and Industry 5.0 are changing healthcare systems through digital transformation and intelligent technologies. I agree with your point that while digital systems improve efficiency and healthcare delivery, they also increase dependency on technology, which creates major risks when systems fail.

One important issue highlighted in your post is the balance between innovation and resilience. Healthcare organisations are rapidly adopting AI, cloud systems, and digital patient records to improve operational efficiency. However, as you mentione[illegible] systems must also be reliable and secure because healthcare environments involve sensitive information and life-critical se[illegible] According to Heeks (2020), digital transformation in healthcare can only succeed when organisations combine technology adoption with strong governance and risk management practices.

I also think your Industry 5.0 perspective is very relevant because it shifts the focus from only automation toward human wellbeing and sustainability. One practical solution could be implementing stronger backup systems and hybrid workflows where critical services can continue manually if IT systems become unavailable. Regular cybersecurity testing and employee training would also help reduce operational risks (ENISA, 2021).

Another important point is public confidence. When patients lose trust in digital healthcare systems, it can affect the adoption of future innovations such as AI-supported diagnostics or remote healthcare services. Building resilient infrastructure and transparent communication strategies during system failures can help organisations maintain public trust and improve long-term digital adoption (Metcalf, 2024).

This post effectively shows that Industry 5.0 is not only about advanced technology but also about creating systems that are secure, reliable, and centred around human needs. The NHS outage demonstrates why resilience and preparedness must become priorities alongside digital innovation.

References

ENISA (2021) *Cybersecurity and Resilience in Healthcare Systems*. European Union Agency for Cybersecurity.

Heeks, R. (2020) *Information and Communication Technology for Development*. London: Routledge.

Metcalf, G.S. (2024) 'An Introduction to Industry 5.0: History, Foundations, and Futures', in Nousala, S., Metcalf, G.S. and Ing, D. (eds.) *Industry 4.0 to Industry 5.0*. Singapore: Springer.

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Peer Response

by Sonya Jackson - Monday, 11 May 2026, 1:30 PM

Hi Sara,

I read with interest your post highlighting the NHS IT outage that occurred in 2022. As you highlighted, the impact of such events can extend beyond financial loss, impacting public health, safety and confidence in health care systems.

Other colleagues also highlighted examples of failures of information systems within healthcare and the impact of these. My thoughts turned to the Increasing digitalisation and centralisation of healthcare services, and how this can bring opportunity for optimisation of healthcare management, improved patient journeys, enhanced research capabilities and greater efficiency within the health care service, but at the same time, increases exposure to, and impact of cyber security threats, including ransomware attacks, misuse of highly sensitive data and potential espionage risks associated with development of large interconnected systems.

Whilst plans for a federated platform may mitigate some concerns, it does not eliminate these risks entirely (National Cyber Security Center 2025). Proposals for the involvement of large third-party companies in management of healthcare data has generated significant public concern, with research indicating that patients may withdraw consent for data sharing due to privacy concerns, potentially impacting the completeness and usefulness of such resources for both research and healthcare planning (Murgia and Neville, 2026; Campbell, 2023)

As the NHS continues to progress towards integration and interoperability of patient data, robust cyber security, risk management and protection of patient confidentiality are key, and remain essential to maintaining both operational resilience and public trust (NHS England, n.d)

References

Campbell, D. (2023) 'Patients may shun new NHS data store over privacy fears, doctors warn', The Guardian, 10 November. Available at: The Guardian article (Accessed: 11 May 2026).

National Cyber Security Centre (2025) Principle 5: Avoid putting too much sensitive data together, in Security principles for protecting the most sensitive personal information in datasets. Available at: https://www.ncsc.gov.uk/collection/security-principles-protecting-most-sensitive-personal-information-in-datasets/principle-5-avoid-putting-too-much-sensitive-data-together (Accessed: 11 May 2026).

NHS England (n.d.) Commissioning flows. Available at: https://www.england.nhs.uk/data-services/commissioning-flows/ (Accessed: 11 May 2026).

Murgia, M. and Neville, S. (2026) 'NHS to grant Palantir contractors "unlimited access" to patient data', Financial Times, 11 May.

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